ORIGINAL

MODULE: SIC
MEMORY: *AC* [18]; *MD* [18]; *PC* [13]; *IR* [13]; *IA* [13]; *IB* [13]; *MA* [13]; lf; M [8193, 18]
INPUTS: start
BUSES: ABUS [18]; BBUS [18]; OBUS [19]

1. $\rightarrow (\overline{start})/(1)$
2. $MA \leftarrow PC$
3. $MD \leftarrow \mathrm{BUSFN}(\mathrm{M}; \mathrm{DCD}(MA))$
4. $IR \leftarrow MD$
5. $\rightarrow (IR_0 \wedge IR_1 \wedge IR_2)/(25)$
6. NO DELAY
   $\rightarrow ((\overline{IR_3} \wedge \overline{IR_4}), (\overline{IR_3} \wedge IR_4), IR_3)/(13,7,10)$
7. $MA \leftarrow \mathrm{IR}_{5:17}$
8. $MD \leftarrow \mathrm{BUSFN}(\mathrm{M}; \mathrm{DCD}(MA))$
9. $IR_{5:17} \leftarrow \mathrm{MD}_{5:17}$
   $\rightarrow (13)$
10. NO DELAY
    $\rightarrow (IR_4)/(12)$
11. $IR_{5:17} \leftarrow \mathrm{ADD}(IR_{5:17}; IA)$
    $\rightarrow (13)$
12. $IR_{5:17} \leftarrow \mathrm{ADD}(IR_{5:17}; IB)$
13. NO DELAY
    $\rightarrow (\overline{IR_0 \wedge IR_1})/(15)$
14. $PC \leftarrow IR_{5:17}$
    $\rightarrow (2)$
15. $MA \leftarrow IR_{5:17}$
    $\rightarrow (IR_0)/(21)$
16. $MD \leftarrow \mathrm{BUSFN}(\mathrm{M}; \mathrm{DCD}(MA))$
    $\rightarrow (\overline{IR_1} \wedge \overline{IR_2})/(18)$
17. $AC \leftarrow \Big(MD!\,(MD \wedge AC)!\,\big(\mathrm{ADD}_{1:18}(MD; AC)\big)\Big) * \big((\overline{IR_1} \wedge IR_2), (IR_1 \wedge \overline{IR_2}), (IR_1 \wedge IR_2)\big)$
    $lf * (IR_1 \wedge IR_2) \leftarrow \mathrm{ADD}_0(MD; AC)$
    $\rightarrow (24)$
18. $MD \leftarrow \mathrm{INC}(MD)$
19. $M * \mathrm{DCD}(MA) \leftarrow MD$
    $\rightarrow (\vee/MD)/(24)$
20. $PC \leftarrow \mathrm{INC}(PC)$
    $\rightarrow (24)$
21. $MD \leftarrow \Big((AC)!\,\big(5 \uparrow 0, \mathrm{INC}(PC)\big)\Big) * (IR_2, \overline{IR_2})$
22. $M * \mathrm{DCD}(MA) \leftarrow MD$
    $\rightarrow (IR_2)/(24)$
23. $PC \leftarrow IR_{5:17}$
24. $PC \leftarrow \mathrm{INC}(PC)$
    $\rightarrow (2)$
25. NO DELAY
    $\rightarrow (IR_3)/(50)$
26. NO DELAY
    $\rightarrow (IR_5)/(30)$

27. NO DELAY
$\rightarrow ((IR_6 \wedge IR_7), (\overline{IR_6 \wedge IR_7}))/(1, 29)$

(No Step 28)

29. $AC * (IR_8 \vee IR_9) \leftarrow ((\overline{18 \uparrow 0}))!\,(18 \uparrow 0)!\ \overline{AC})$
$* ((\overline{IR_8} \wedge IR_9), (IR_8 \wedge \overline{IR_9}), (IR_8 \wedge IR_9));$
$If * IR_6 \leftarrow 0; If * IR_7 \leftarrow 1$
$\rightarrow (33)$

30. $\rightarrow (IR_4)/(32)$ NO DELAY

31. $If, AC \leftarrow AC, If$
$\rightarrow (33)$

32. $If, AC \leftarrow AC_{17}, If, AC_{0:16}$

33. NO DELAY
$\rightarrow (IR_{10})/(40)$

34. NO DELAY
$\rightarrow ((\overline{IR_{11}} \wedge \overline{IR_{12}}), (\overline{IR_{11}} \wedge IR_{12}), (IR_{11} \wedge \overline{IR_{12}}),$
$(IR_{11} \wedge IR_{12}))/(35, 37, 38, 39)$

35. NO DELAY
$\rightarrow (\overline{IR_{13} \wedge \overline{If}})/(43)$

36. $PC \leftarrow \mathrm{INC}(PC)$
$\rightarrow (24)$

37. $AC \leftarrow (5 \uparrow 0, IA\,!\ 5 \uparrow 0, IB) * (\overline{IR_{13}}, IR_{13})$
$\rightarrow (43)$

38. $IA \leftarrow ((AC_{5:17})\,!\,(\mathrm{INC}(IA))) * (\overline{IR_{13}}, IR_{13})$
$\rightarrow (43)$

39. $IB \leftarrow ((AC_{5:17})\,!\,(\mathrm{INC}(IB))) * (\overline{IR_{13}}, IR_{13})$
$\rightarrow (43)$

40. NO DELAY
$\rightarrow (IR_4)/(42)$

41. $If, AC \leftarrow AC, If$
$\rightarrow (43)$

42. $If, AC \leftarrow AC_{17}, If, AC_{0:16}$

43. NO DELAY
$\rightarrow (IR_{14})/(45)$

44. $\rightarrow (f, \overline{f})/(36, 24)$

45. NO DELAY
$\rightarrow (IR_4)/(47)$

46. $If, AC \leftarrow AC, If$
$\rightarrow (24)$

47. $If, AC \leftarrow AC_{17}, If, AC_{0:16}$
$\rightarrow (24)$

END SEQUENCE

$f = (AC_0 \wedge IR_{15}) \vee ((\overline{\vee/AC}) \wedge IR_{16}) \vee (\overline{AC_0} \wedge (\vee/AC) \wedge IR_{17})$

END.

EXPLICACION

MODULE: SIC
MEMORY: *AC* [18]; *MD* [18]; *PC* [13]; *IR* [13]; *IA* [13]; *IB* [13]; *MA* [13]; lf; M [8193, 18]
INPUTS: start
BUSES: ABUS [18]; BBUS [18]; OBUS [19]

1. $\rightarrow (\overline{start})/(1)$

INICIO
2. $MA \leftarrow PC$
3. $MD \leftarrow \mathrm{BUSFN}(\mathrm{M}; \mathrm{DCD}(MA))$
4. $IR \leftarrow MD$

5. $\rightarrow (IR_0 \wedge IR_1 \wedge IR_2)/(25)$ ←Excluye Ins. Operativas (111)
6. NO DELAY
$\rightarrow (\underbrace{(\overline{IR_3} \wedge \overline{IR_4})}_{\text{DIR.}}, \underbrace{(\overline{IR_3} \wedge IR_4)}_{\text{INDIR.}}, \underbrace{IR_3}_{\text{INDEX.}})/(13,7,10)$

INDIRECTO
7. $MA \leftarrow \mathrm{IR}_{5:17}$ ←Ponemos dirección indirecta
8. $MD \leftarrow \mathrm{BUSFN}(\mathrm{M}; \mathrm{DCD}(MA))$ ←Sacamos contenido
9. $IR_{5:17} \leftarrow \mathrm{MD}_{5:17}$ ←Obtenemos dirección real
$\rightarrow (13)$

INDEXADO
10. NO DELAY
$\rightarrow (IR_4)/(12)$ ←Elige registro índice
11. $IR_{5:17} \leftarrow \mathrm{ADD}(IR_{5:17}; IA)$ ←Sumamos IA a la dirección actual
$\rightarrow (13)$
12. $IR_{5:17} \leftarrow \mathrm{ADD}(IR_{5:17}; IB)$ ←Sumamos IB a la dirección actual

13. NO DELAY
$\rightarrow (\overline{IR_0 \wedge IR_1})/(15)$ } Excluye JMP (110)

JMP
14. $PC \leftarrow IR_{5:17}$ ←Carga la posición del salto en puntero
$\rightarrow (2)$ ←Vuelve al inicio

DIRECCION
15. $MA \leftarrow IR_{5:17}$ ←Ponemos la dirección final
$\rightarrow (IR_0)/(21)$ ←Bifurcación para escritura y lectura IR0=0 Lectura IR0=1 Escritura

LEER DATO
16. $MD \leftarrow \mathrm{BUSFN}(\mathrm{M}; \mathrm{DCD}(MA))$ ←Leemos el contenido de la dirección final
$\rightarrow (\overline{IR_1} \wedge \overline{IR_2})/(18)$ ←Excluye ISZ (100)

OPERACIÓN "ALU"
17. $AC \leftarrow \Big(\overbrace{MD!\,(MD \wedge AC)}^{\text{LAC}}!\,\overbrace{(\mathrm{ADD}_{1:18}(MD; AC))}^{\text{AND}}\Big)\overbrace{)}^{\text{TAD}} * \big((\overline{IR_1} \wedge IR_2), (IR_1 \wedge \overline{IR_2}), (IR_1 \wedge IR_2)\big)$
$lf * (IR_1 \wedge IR_2) \leftarrow \mathrm{ADD}_0(MD; AC)$ ←Actualizar LF con TAD
$\rightarrow (24)$

ISZ
18. $MD \leftarrow \mathrm{INC}(MD)$ ←Incrementar el contenido
19. $M * \mathrm{DCD}(MA) \leftarrow MD$ ←Guarda el nuevo contenido
$\rightarrow (\vee/MD)/(24)$ ←Verifica si es igual a 0 y si no va al paso 24 para finalizar
20. $PC \leftarrow \mathrm{INC}(PC)$ ←Si es 0 incrementa el puntero y luego va al paso 24 para terminar
$\rightarrow (24)$

ESCRITURA
21. $MD \leftarrow \Big(\overbrace{(AC)!}^{\text{DAC}}\,\overbrace{(5 \top 0, \mathrm{INC}(PC))}^{\text{JMS}}\Big) * (IR_2, \overline{IR_2})$
22. $M * \mathrm{DCD}(MA) \leftarrow MD$ ←Escribimos en la memoria
$\rightarrow (IR_2)/(24)$ ←Bifurcación para el nuevo puntero de JMS
23. $PC \leftarrow IR_{5:17}$ ←Carga de la posición de salto del puntero
24. $PC \leftarrow \mathrm{INC}(PC)$ ←Bifurcación para el nuevo puntero de JMS
$\rightarrow (2)$

IR0	IR1	IR2	NEMONICO
0	0	0	ISZ
0	0	1	LAC
0	1	0	AND
0	1	1	TAD
1	0	0	JMS
1	0	1	DAC
1	1	0	JMP
1	1	1	OPERATIVAS

EXCLUSION I/O

25. NO DELAY
$\rightarrow (IR_3)/(50)$

PRIMER EVENTO

DECISION ROTACION

26. NO DELAY
$\rightarrow (IR_5)/(30)$ ←Rotar IR5=1 o NO rotar IR5=0

CASO NOP

27. NO DELAY
$\rightarrow (\underbrace{(IR_6 \wedge IR_7)}_{\text{HLT}}, (\overline{IR_6 \wedge IR_7}))/(1, 29)$

(No Step 28)

AJUSTE ACUMULADOR

29. $AC * (IR_8 \vee IR_9) \leftarrow (\overbrace{(\overline{18 \top 0})}^{\text{STA}})! \overbrace{(18 \top 0)}^{\text{CLA}}! \overbrace{\overline{AC}}^{\text{CMA}})$
$* ((\overline{IR_8} \wedge IR_9), (IR_8 \wedge \overline{IR_9}), (IR_8 \wedge IR_9));$
$\overbrace{If * IR_6 \leftarrow 0}^{\text{CLL}}; \overbrace{If * IR_7 \leftarrow 1}^{\text{STL}}$
$\rightarrow (33)$

ROTACIONES

30. $\rightarrow (IR_4)/(32)$ NO DELAY
31. $If, AC \leftarrow AC, If$
$\rightarrow (33)$
32. $If, AC \leftarrow AC_{17}, If, AC_{0:16}$

SEGUNDO EVENTO

DECISION ROTACION

33. NO DELAY
$\rightarrow (IR_{10})/(40)$ ←Rotar IR10=1 o NO rotar IR10=0

34. NO DELAY
$\rightarrow ((\overbrace{\overline{IR_{11}} \wedge \overline{IR_{12}}}^{\text{NOP/SZL}}), (\overbrace{\overline{IR_{11}} \wedge IR_{12}}^{\text{DFA/DFB}}), (\overbrace{IR_{11} \wedge \overline{IR_{12}}}^{\text{DTA/INA}}), (\overbrace{IR_{11} \wedge IR_{12}}^{\text{DTB/INB}}))/(35, 37, 38, 39)$

NOP/SZL

35. NO DELAY
$\rightarrow (\overline{IR_{13} \wedge \overline{If}})/(43)$ ←Verifica si LF=0 sino va al siguiente evento
36. $PC \leftarrow \text{INC}(PC)$ ←Si es 0 incrementa el puntero y luego va al paso 24 para terminar
$\rightarrow (24)$

DFA/DFB

37. $AC \leftarrow (5 \top 0, IA ! 5 \top 0, IB) * \overbrace{(\overline{IR_{13}}, IR_{13})}^{\text{DFA/DFB}}$
$\rightarrow (43)$

DTA/INA

38. $IA \leftarrow ((AC_{5:17}) ! (\text{INC}(IA))) * \overbrace{(\overline{IR_{13}}, IR_{13})}^{\text{DTA/INA}}$
$\rightarrow (43)$

DTB/INB

39. $IB \leftarrow ((AC_{5:17}) ! (\text{INC}(IB))) * \overbrace{(\overline{IR_{13}}, IR_{13})}^{\text{DTB/INB}}$
$\rightarrow (43)$

ROTACIONES

40. NO DELAY
$\rightarrow (IR_4)/(42)$
41. $If, AC \leftarrow AC, If$
$\rightarrow (43)$
42. $If, AC \leftarrow AC_{17}, If, AC_{0:16}$

TERCER EVENTO

DECISION ROTACION

43. NO DELAY
$\rightarrow (IR_{14})/(45)$

SALTOS

44. $\rightarrow (f, \overline{f})/(36, 24)$

ROTACIONES
- 45. NO DELAY
  $\rightarrow (IR_4)/(47)$
- 46. $If, AC \leftarrow AC, If$
  $\rightarrow (24)$
- 47. $If, AC \leftarrow AC_{17}, If, AC_{0:16}$
  $\rightarrow (24)$

END SEQUENCE

$$f = \underbrace{(AC_0 \wedge IR_{15})}_{AC < 0} \vee (\underbrace{(\overline{\vee/AC}) \wedge IR_{16}}_{AC = 0}) \vee (\underbrace{\overline{AC_0} \wedge (\vee/AC) \wedge IR_{17}}_{AC > 0})$$

END.